

# Quick Check

Lesson 4: Germination: Conditions Necessary and Types (Hypogeal vs Epigeal) in Maize H614 and Rosecoco Beans

*7 multiple-choice questions • choose A, B, C or D*

Sub-Strand 2.2: Growth and Development in Plants | Lesson 4

**Mr. Wafula's healthy, non-dormant maize and bean seeds were planted in the same field after the same rain, but some still failed to germinate. Which set lists the conditions a healthy, non-dormant seed needs in order to germinate?**

**A** Water, oxygen and a suitable temperature

**B** Oxygen, soil and fertiliser

**C** Light, water and soil

**D** Water, light and a suitable temperature

Fatuma set up four jars, each with 20 Rosecoco bean seeds: water + air + 25 °C; dry; no air (boiled cooled water with an oil layer); and wet but kept at 4 °C. Which of these was kept the SAME in all four jars so that the test was fair?

- A The oxygen supply
- B The temperature
- C The water supply
- D The number of seeds (20) and the seed variety

**A group is counting germinated Rosecoco bean seeds in a jar on Day 5. When should a seed be counted as GERMINATED?**

- A** When the first leaves open
- B** When the seed coat becomes soft and swollen
- C** When the radicle has come out of the seed
- D** When the shoot appears above the soil

In Fatuma's set-up, the jar with water at 25 °C but no air (boiled cooled water with an oil layer) had only 1 of 20 seeds germinate. In the jar with water, air and 25 °C, 18 of 20 germinated. What does this comparison show?

- A** Water is enough on its own for germination
- B** Oxygen is needed, because the embryo uses it in aerobic respiration to release energy for growth
- C** Light is needed, because the oil layer blocked light
- D** Oxygen is not needed, because one seed still germinated

**On Mr. Wafula's farm, 6 bean seeds and 3 maize seeds failed to germinate in a low patch that was waterlogged after 25 mm of rain. What is the best explanation?**

- A** The rain washed away the light the seeds needed
- B** Water filled the soil pores, so the seeds got too little oxygen
- C** The soil became too cold for any seed to absorb water
- D** The seeds became dormant because they had too much water

**Rosecoco beans show epigeal germination, and the cotyledons are carried above the soil. Which part of the seedling makes this happen?**

- A** The scutellum grows upwards out of the soil
- B** The coleoptile pierces the soil and lifts the cotyledons
- C** The hypocotyl elongates as an arch and pulls the cotyledons above the soil
- D** The radicle grows upwards to push the cotyledons out

In the time-lapse results, the bean radicle came out on Day 3 and the bean shoot appeared above the soil on Day 8 (emergence). How many days passed between germination and emergence in the bean?

A 5 days

B 3 days

C 8 days

D 11 days

# All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).